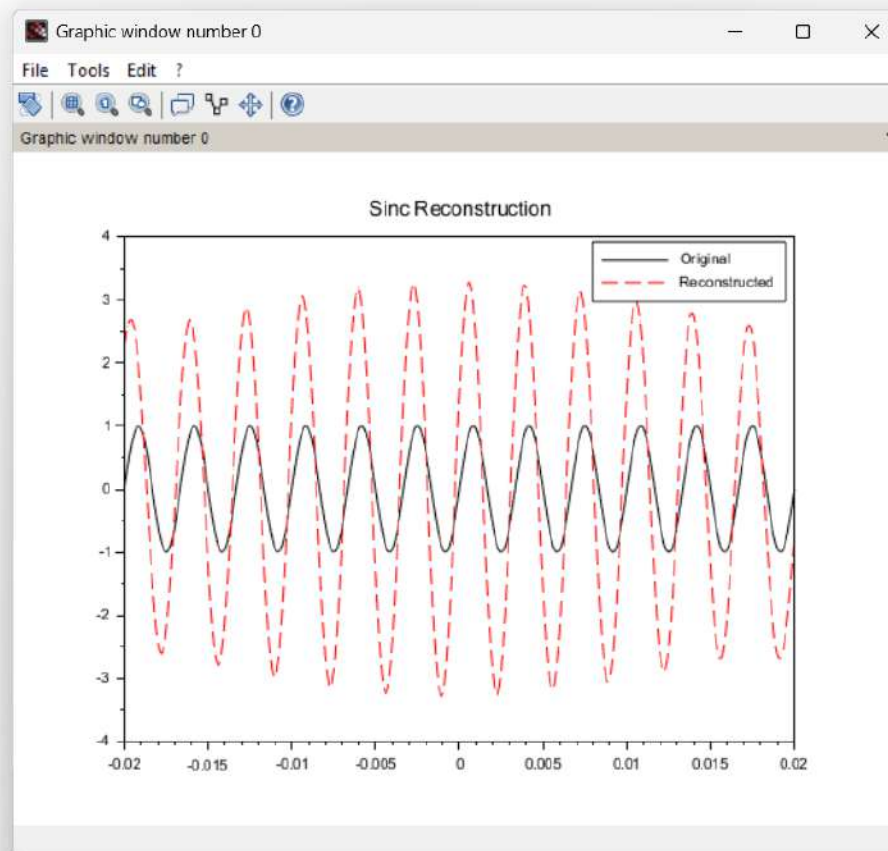


```
1 function y = sinc_interp(x, t, Ts)
2 y = 0*t;
3 for k = 1:length(x)
4 y = y + x(k) * sinc((t - (k-1) * Ts) / Ts);
5 end
6 endfunction
7 t = -0.02:1e-4:0.02;
8 f0 = 300;
9 x = sin(2*pi*f0*t);
10 fs = 2000; Ts = 1/fs;
11 n = -40:40;
12 ns = n*Ts;
13 xs = sin(2*pi*f0*ns);
14 x_rec = sinc_interp(xs, t, Ts);
15 clf;
16 plot(t, x, 'k');
17 plot(t, x_rec, 'r--');
18 legend("Original", "Reconstructed");
19 xtitle("Sinc-Reconstruction");
20
```





EXP_7.3 SS.scil (E:\EXP_7.3 SS.scil) - SciNotes

EXP_7.3 SS.scil

```
1 // Time vector.
2 t = -1:0.001:1; // fs = 1000 Hz.
3 x = exp(-t.^2); // Gaussian pulse.
4 // FFT length (must be >= length(x)).
5 N = 2048; // choose bigger than length(x).
6 Xf = fft([x zeros(1, N - length(x))]); // zero-padding to N.
7 // Frequency vector.
8 fs = 1000; // Hz.
9 f = (-N/2:N/2-1) * (fs/N);
10 // Magnitude spectrum.
11 magX = abs(fftshift(Xf));
12 // Phase spectrum.
13 phaseX = atan(imag(fftshift(Xf)), real(fftshift(Xf)));
14 // Plot.
15 clf;
16 subplot(2,1,1);
17 plot(f, magX);
18 xtitle("Magnitude Spectrum", "Frequency (Hz)", "Magnitude");
19 subplot(2,1,2);
20 plot(f, phaseX);
21 xtitle("Phase Spectrum", "Frequency (Hz)", "Phase (rad)");
22
```

